Homework — 2.1.4 Thinking Logically

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 2.1.4 lesson.

Part A — This week's topic (~14 marks)

1. A theme-park ride allows a rider on only if they are **at least 1.4 m tall**.

(a) Write the exact Boolean **condition** for the decision point. **[1]** *(AO2)*

(b) State the outcome of the TRUE branch and the FALSE branch. **[2]** *(AO2)*

2. A bank lets a customer withdraw from a cash machine only if the amount requested is **no more than** the balance **and** no more than the daily limit.

Write a suitable Boolean condition using a logical operator, and state what happens when the condition is **false**. **[2]** *(AO2)*

3. A water-tank controller turns a pump **ON** when the level falls below 20%, and **OFF** when it rises above 80%; otherwise it leaves the pump unchanged.

(a) Write the two Boolean conditions used at these decision points. **[2]** *(AO2)*

(b) State how many possible outcomes (including "leave unchanged") this logic has. **[1]** *(AO2)*

4. A shop gives a loyalty discount when a customer has spent **£100 or more**. A student writes `IF spent > 100`.

State whether a customer who has spent exactly £100 receives the discount under this condition, and rewrite the condition so that exactly £100 *does* qualify. **[2]** *(AO2)*

5. An online exam system decides a candidate's grade band. If the score is **70 or above** it awards a Distinction; otherwise, if it is **40 or above** it awards a Pass; otherwise it awards a Fail.

Using *thinking logically*, identify the decision points, write the exact Boolean conditions in the correct order, and state the outcome of each branch. **[4]** *(AO2)*

Part B — Retrieval: keep it fresh (~6 marks)

6. *(2.1.3)* In a payroll program the order is: read hours → calculate gross pay → deduct tax → output payslip. State **why** "deduct tax" must come after "calculate gross pay". **[2]** *(AO2)*

7. *(1.3.2)* State what is meant by a **primary key** and a **foreign key** in a relational database. **[2]** *(AO1)*

8. *(1.4.1)* Convert the denary number **45** into **8-bit binary**. **[2]** *(AO2)*